Ministry of High Education
Modern Academy for Computer Science and Management Technology
Department: Computer Science
Program: Computer Science
Time: 1 Hour
Examiner: (Dr Ahmed Ibrahim, Dr. Lamiaa Hassaan)

Academic Year: 2022-2023
Term: Summer Final Term (credit)
Subject: Data Structures (2022)
Code: CS201
Specialization: 2nd Level CS
Aids: None
Total Marks: 60 Marks

Number of Questions: 4 | **Questions for Final Examination** | **Number of pages: 1**

Answer the following questions: **(15 Marks for each)**

First Question: You have the matrix which represents the degrees of 6 students

Name	Subject 1	Subject 2	Subject 3	Subject 4
Ahmed	70	97	50	75
Mohamed	60	90	80	79
Halaa	50	45.5	60	81
Dina	80	80	70	60
Abas	40	70	75	55
Hoda	75	60.5	85	50

Using array of struct write a program to:

- Calculate the average value of each subject.
- Calculate the total degree of each student.

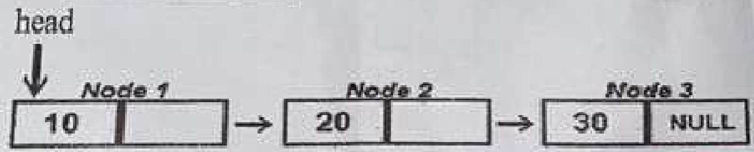
Second Question: You have the Linked-Queue, **write a program to:**

- Remove an item from the Queue.
- Insert a new item which value = 50

Third Question: You have the Linked-Stack, **write a program to:**

- Insert a new two nodes with values = 50 and 70.
- Remove a node from the stack.

Fourth Question: Write a program to insert a new node in its order of this linked list?



With my best wishes

إدارة التخطيط والإمتحانات
مطبعة الإمتحانات

Data Structures

السؤال الاول

```cpp
// اذكر الله
// صل على خير خلق الله محمد

// الكومنتات اللى مكتوبة فى الكود للتوضيح فقط ولست مطالب بكتابتها فى ورقة الامتحان

#include <iostream>
using namespace std;

int main()
{
    // define struct of student data
    struct student
    {
        string name;
        // array to get degrees of all 4 subjects
        float degree[4];
        // varibal total to get student total degrees
        float total = 0;
    };
    // define array of students;
    student students_data[6];
    // array to calcoulate subjects average over all students
    float avg[4] = {0};
    // take student data from user
    for (int i = 0; i < 6; i++)
    {
        cout << "______________________________________________________________\n"; // "سطر لتحسين شكل الاوتبوت فقط "ملوش لازمة
        // take student name
        cout << "please, enter data of student number " << i + 1 << '\n';
        cout << "student name                ";
        cin >> students_data[i].name;
        // take student degrees
        for (int j = 0; j < 4; j++)
        {
            float temp;
            cout << "degree of subject number " << j + 1 << "    ";
            cin >> temp;
            students_data[i].degree[j] = temp;
            // add subject degree to calc student total
            students_data[i].total += temp;
            // add subject degree to calc subject avrage
            avg[j] += temp;
        }
    }
    // for loop to calc final average result
    for (size_t i = 0; i < 4; i++)
        avg[i] /= 6;
    // print every students total degrees
    cout << "______________________________________________________________\n"; // "سطر لتحسين شكل الاوتبوت فقط "ملوش لازمة
    for (int i = 0; i < 6; i++)
        cout << "total of student number " << i + 1 << " equql to " << students_data[i].total << '\n';
    // print every subject average
    cout << "______________________________________________________________\n"; // "سطر لتحسين شكل الاوتبوت فقط "ملوش لازمة
    for (int i = 0; i < 4; i++)
        cout << "average of degree number " << i + 1 << "equal to " << avg[i] << '\n';
    // end of the program
    return 0;
}
```

السؤال الثانى

```
// اذكر الله
// صلِ على خير خلق الله محمد

// الكومنتات اللى مكتوبة فى الكود للتوضيح فقط ولست مطالب بكتابتها فى ورقة الامتحان

#include <iostream>
using namespace std;

struct linked_Queue
{
    // rest of my linked queue functions
    // Enqueue (add) an element to the linked queue
    void inQueue(int element)
    {
        node *temp = new node();
        temp->data = element;
        if (last == NULL)
        {
            // If the queue is empty, set both head and last to the new node
            head = temp;
            last = temp;
        }
        else
        {
            // Add the new node to the end of the queue and update last
            last->next = temp;
            last = temp;
        }
    }

    // Dequeue (remove) an element from the linked queue
    void deQueue()
    {
        if (head == NULL)
        {
            cout << "Queue is empty\n";
            return;
        }
        if (head->next == NULL)
        {
            // If there's only one element, delete it and set head and last to NULL
            delete head;
            head = NULL;
            last = NULL;
        }
        else
        {
            // Remove the front node and update head
            node *temp = head;
            head = head->next;
            delete temp;
        }
    }
};

int main()
{
    // create queue
    linked_Queue queue;
    // remove an element
    queue.deQueue();
    // add 50 to queue
    queue.inQueue(50);
    return 0;
}
```

السؤال الثالث

```cpp
// اذكر الله
// صلِ على خير خلق الله محمد

// الكومنتات اللى مكتوبة فى الكود للتوضيح فقط ولست مطالب بكتابتها فى ورقة الامتحان

#include <iostream>
using namespace std;

struct linked_Stack
{
    // rest of my linked stack functions
    // Insert (push) an element to the linked stack
    void insert(int element)
    {
        node *temp = new node();
        temp->data=element;
        temp->next = head;
        head = temp;
    }

    // Remove (pop) an element from the linked stack
    void remove()
    {
        if (head == NULL)
        {
            cout << "Stack is empty\n";
            return;
        }
        node *temp = head;
        head = head->next;
        delete temp;
    }
};

int main()
{
    // create queue
    linked_Stack stack;
    // insert a new 2 nodes with valuses 50 and 70
    stack.insert(50);
    stack.insert(70);
    // remove a node from the stack
    stack.remove();
    return 0;
}
```

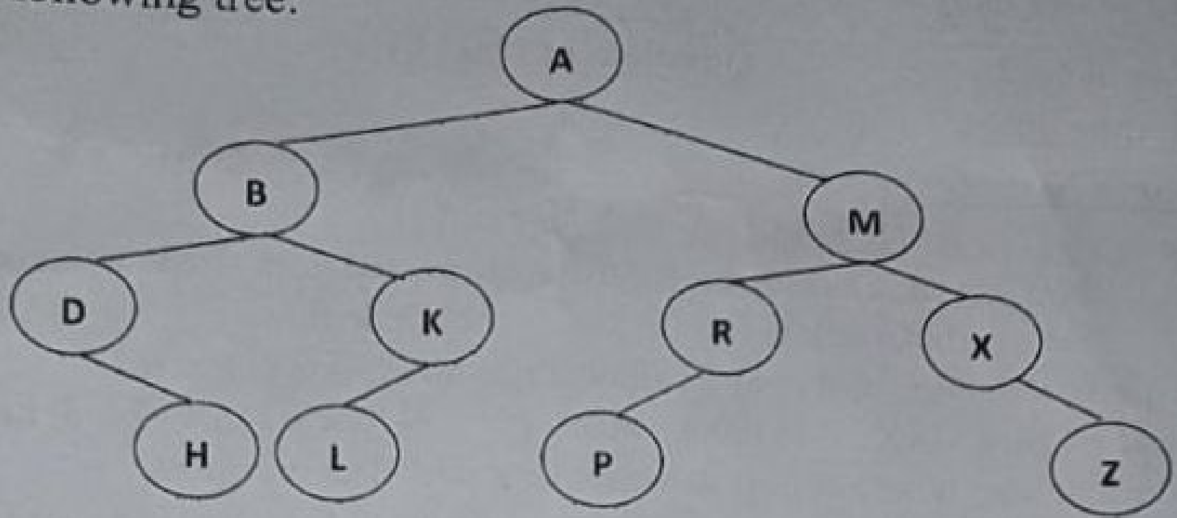
السؤال الرابع

```
// اذكر الله
// صلِ على خير خلق الله محمد

// الكومنتات اللى مكتوبة فى الكود للتوضيح فقط ولست مطالب بكتابتها فى ورقة الامتحان

#include <iostream>
using namespace std;

struct linkedlist
{
    // rest of my linked list functions
    // Insert (push) an element to the linked list in it's order
    void insert_in_order(int element)
    {
        node *temp = new node();
        temp->data = element;
        // check the 2 conditions to add it in begaining
        // 1st : linked list is empty
        // 2nd : the first element in my linked list is greater than or equal the node i need to add
        if (is_empty() || head->data >= element)
        {
            // insert_at_beginning(element);
            temp->next = head; // 1st step
            head = temp;       // 2nd step
            return;
        }
        // find the targeted possition have 2 condition
        // 1st : if i reach the end it''s mean all of the list element is less targeted node and i will add it in the end
        // 2nd : find the first element is greater than targeted node and add it before it
        node *cur = head;
        while (cur->next != NULL && cur->next->data < element)
            cur = cur->next;
        // add node in the targeted position
        temp->next = cur->next;
        cur->next = temp;
    }
};

int main()
{
    // create queue
    linkedlist list;
    // insert a new 3 nodes with valuses 10 ,20 ,and 30
    list.insert_in_order(10);
    list.insert_in_order(20);
    list.insert_in_order(30);
    return 0;
}
```

Question Two: Answer ONLY FIVE of the following: (40 marks – 8 for each)

a- Store the values **14, 17, 25, 37, 34, 16** using division method. Apply **linear probe** and **quadratic probe** in collision situations. **Array size = 11**

b- Consider the following tree:



(i) Write the output of inorder and postorder traversals

(ii) Write C++ function of preorder traverse

c- Consider a circular queue of size = 5, show the <u>output</u> of the following code segment:

```
void main( )
{    enqueue (10);      enqueue (20);
     cout<<"Head = "<<head <<"Tail = "<<tail<<"\n";
     enqueue (30);
     cout<<dequeue( )<<"\n";
     int x = dequeue ( );
     cout<<"Head = "<<head<<"Tail = "<<tail<<"\n";
}
```

d- You have a Linked Queue, write c++ functions to:

- remove an item from the queue.
- insert a new item which value = 50

e- You have a Doubly Linked List, write C++ function to insert a new node in its order.

f- You have a Linked Stack, write C++ program to remove all the items from the stack.

Ministry of Higher Education
Modern Academy For Computer Science and Management Technology
Department: Computer Science
Program: Computer Science
Time: 2 Hours
Examiners: Dr. Ahmed Ibrahim - Dr. Lamia Hassaan

Academic Year: 2022-2023
Term: 1st Term
Subject: Data Structures
Code: CS201 (CREDIT 2022)
Specialization: Computer Science–2nd level
Aids: None
Total Marks: 60 Marks

Number of Questions: 2 | Questions of Final Exam | Number of pages: 2

Answer ALL questions:

Question One: (20 marks)

I- Choose the correct answer (10 marks – 1 mark for each)

1- Pushing an element into stack already having five elements and stack size is 5, this condition is called.......
a) Crash b) Overflow c) Underflow d) User flow

2- In a linear queue, if implemented using an array of size MAX, gets full when?
a) Front= (rear + 1)mod MAX b) Front = rear + 1 c) Rear = MAX– 1 d) Rear = front

3- Which of the following is not a correct statement for a stack data structure?
a)Stack follows FIFO b) Arrays can be used to implement the stack c) Elements are stored in a sequential manner d) Top of the stack has the last inserted element

4- Which of the following data structure is non-linear type?
a) Linked List b) Stack c) Tree d) Circular queue

5- Which of the following determines the need for the circular queue?
a) Avoid memory wastage b) Access queue using priority c) Follows the FIFO principle d) None

6- If a Binary Search Tree is traversed in the order *right, root, left* recursively, the output sequence will be in:
a) ascending order b) descending order c) no specific order d) *right, root, left* cannot be done

7- If circular queue is implemented using array having size MAX_SIZE, which of the following conditions used to specify that the circular queue is empty?
a) front = rear = 0 b) front = rear + 1 c) front = rear = -1 d) none

8- When the pre-order and post order traversal of a Binary Tree generates the same output. Then the tree has:
a) two nodes b) only one node c) three nodes d) any number of nodes

9- To insert an item into a stack, identify the correct set of statements:
a) top++; stack[top]=item; b) item = stack[top]; top --; c) stack[item] = top; item++; d) stack[top] = item; top++;

10- What are the number of nodes of left and right subtree of the binary tree if the data is inserted in the following order: 45, 15, 8, 56, 5, 65, 47, 12, 18, 10, 73, 50, 16, 61
a) 8 nodes to the left and 5 nodes to the right
b) 6 nodes to the left and 7 nodes to the right
c) 7 nodes to the left and 6 nodes to the right
d) 5 nodes to the left and 8 nodes to the right

II- Select True or False (10 Marks – 1 mark for each)

1- The condition if(front == 0 && rear == MAX_SIZE – 1) is enough to check the fullness of a circular queue
2- Hash tables store keys in ordered locations
3- The value of REAR is increased by 1 when an element is inserted in the queue
4- The following is a valid code of class Node of a singly linked list: class Node {int data, Node next;}
5- When a hash function computes the same location for two or more keys, this is a collision situation
6- To print the keys of a binary search tree in an ascending order, you may apply inorder or postorder traversals
7- Two linked lists are needed in order to implement a linked stack.
8- On inserting a new node N in a doubly linked list, N->next = Null is a must
9- Left –Right-Parent represents the Postorder Traversal of a Binary Tree
10- Insertion is done at the front of a linked queue

Ministry of High Education
Modern Academy for Computer Science and Management Technology
Department: Computer Science
Program: Computer Science
Time: 2 Hour
Examiner: (Dr Ahmed Ibrahim, Dr. Lamiaa Hassaan)

Academic Year: 2023-2024
Term: Summer Term
Subject: Data Structures (credit)
Code: CS201
Specialization: 2nd Level CS
Aids: None
Total Marks: 50 Marks

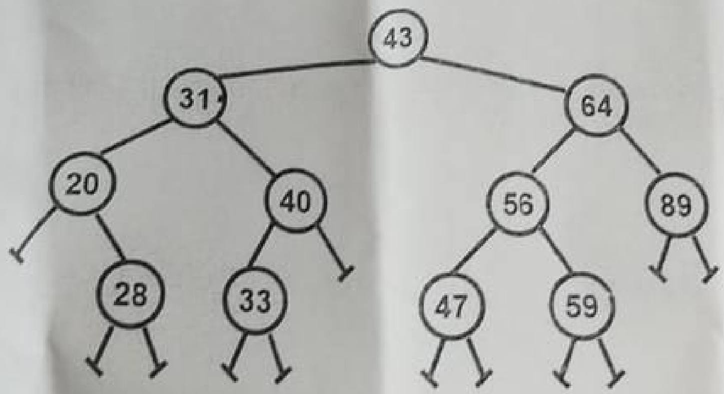
Number of Questions: 6 | Questions for Final Examination | Number of pages: 1

Answer five questions only of the following questions: (10 Marks for each)

1. **Using array of struct write a program to:**
   - take the values of this table from the user.
   - calculate the total salary of each employee after adding the bonus and subtracting the deduction.

Name	Salary	Bonus	Deduction
Dina	1000	100	50
Mohamed	2000	0	0
Abas	2000	100	0
Hoda	3000	0	200

2. **Write a function to insert a new node in its order of a Linked-List?**
3. **You have a Linked-Stack, write a program to:**
   - insert two new nodes with values = 50 and 70.
   - remove a node from the linked-stack.
4. **You have a Linked-Queue write a program to remove all the items of the queue.**
5. **Traverse this tree nodes using Inorder, Preorder, and Postorder.**



6. **Store the values 14, 17, 25, 37, 34, 16 using division method. Apply linear probe in collision situations, where array size = 11.**

With my best wishes

Ministry of High Education
Modern Academy for Computer
Science and Management Technology
Department: Computer Science
Program: Computer Science
Time: 2 Hour
Examiner: (Dr Ahmed Ibrahim, Dr. Lamiaa Hassaan)

Academic Year: 2023-2024
Term: Summer Term (مرجعي)
Subject: Data Structures (credit)
Code: CS201
Specialization: 2nd Level CS
Aids: None
Total Marks: 60 Marks

Number of Questions: 6 **Questions for Final Examination** **Number of pages: 1**

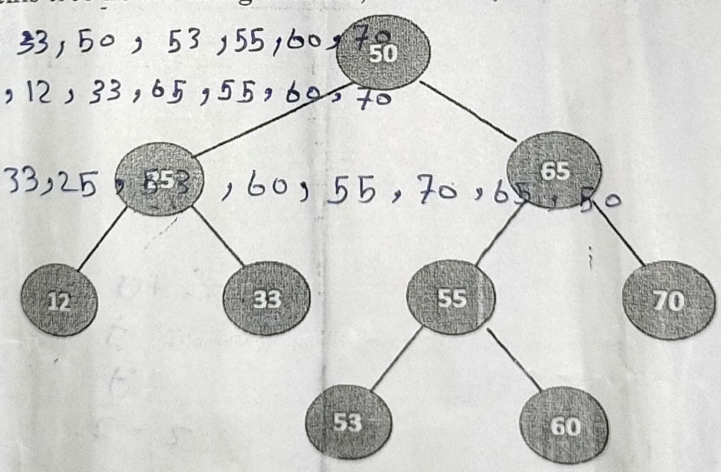
Answer <u>five</u> only of the following questions: **(12 Marks for each)**

1. **Using array of struct write a program to:**
   - take the values of this table from the user.
   - calculate the average value of the subject degrees (sub1, sub2).

Name	Sub1	Sub2
Ahmed	60	70
Mohamed	80	60
Abas	50	40.5
Hoda	70.5	75

2. **Write the functions to remove all the items from a Linked-List.**
3. **You have a Linked-Stack; write a program to remove all the items of the stack.**
4. **You have a Linked-Queue, write a program to:**
   - Insert two new nodes with values = 50 and 70.
   - Remove a node from the queue.
5. **Traverse this tree nodes using Inorder, Preorder, and Postorder.**

inorde: 12, 25, 33, 50, 53, 55, 60, 70
Preorder: 50, 25, 12, 33, 65, 55, 60, 70
postorder: 12, 33, 25, 53, 60, 55, 70, 65, 50



3 6 4 5 0

6. **Store the values 12, 15, 21, 31, 90 using division method. Apply linear probe in collision situations, where array size = 9.**

insert key 12, when i=0, h(12,0) = ((12 % 9)+1) % 9 =

With my best wishes

إدارة التخطيط والامتحانات

Ministry for High Education
Modern Academy for Computer Science and Management Technology
Department: Computer Science
Program: Computer Science
Time: 1 Hrs.
Examiners: Dr. Ahmed Ibrahim Dr. Lamia Hassaan

A

Academic Year: 2023-2024
Term: 1st Mid-Term (Credit)
Subject: Data Structures
Code: CS201
Specialization: 2nd Level CS
Aids: None
Total Marks: 20

Number of Questions: 2	Model Answer	Number of pages: 3

Model Answer A 1/3

Answer the following questions:

First Question: **(5 Marks (1 Mark each))**

Fill in the spaces with suitable word:

a. Removing an element occurs at the **front** of the **Queue**
b. The disadvantage of **array** is **fixed size**
c. Function **push ()** in Stack used to **insert item**

Show which sentence is true and which is false and correct the false sentences:

d. Queue is a data structure which insertions and deletions are restricted to one end. (**False**) **insertions at rear and deletions at front**
e. A Stack, in other words, is called a LIFO list. (**True**)

Second Question: Write C++ code to implement the following functions:
(15 Marks (5 Marks each))

a- **POP** and **IsEmpty** Stack functions.

Answer:

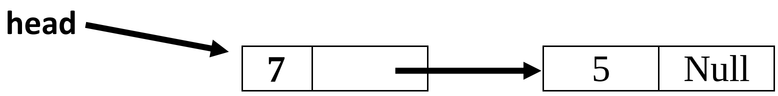
```
//check if stack is empty
bool isEmpty(int top)
{
   if (top == -1) return true;
   else return false;
}
//Pop function
int pop( )
{
if (isEmpty(top)) { cout<<"Stack underflow"; return -1;}
else
{  int item = stack_arr[top];  top = top-1;  return item; }
}
```

b- **Enqueue** and **IsFull** Queue functions.

Answer:

```
bool isFull(int tail)
{
        if ( tail== maxsize-1)
                return true;
        else return false;
}
void enqueue(int item)
        {
                if(isFull(tail))
                        { cout<<"\nQueue is full"; return;}
                else
                        {
                                if ( head == -1 ) head = 0;
                                tail = tail + 1;
                                queue[tail] = item;
                        }
                }
```

c- Write **node class**, **linked list class**, **insertion at the end** function then use them to create the linked list as the following figure (**insert value 7 then value 5**).



Answer:

```
#include <stdafx.h>
#include <iostream>
#include <string>
using namespace std;
class Node
{
public:
        int data;
        Node *next;
        Node()
        {
                data=0;
                next=NULL;
        }
};
class Linkedlist
{
```

Model Answer A 3/3

```
public:
        Node *head;
        Linkedlist()
        {
                head=NULL;
        }

        bool isEmpty()
        {
                if (head==NULL)
                        return true;
                else
                        return false;
        }
        void insertatend(int item)
        {
                Node *newnode=new Node();
                newnode->data=item;
                newnode->next= NULL;
                Node *temp1=head;
                while (temp1->next!=NULL) temp1=temp1->next;
                temp1->next=newnode;
        }
};
void main()
{
        Linkedlist list;
        list.insertatbeginning(7);
        list.insertatbeginning(5);
}
```

With my best wishes

Ministry for High Education
Modern Academy for Computer Science and Management Technology
Department: Computer Science
Program: Computer Science
Time: 1 Hrs.
Examiners: Dr. Ahmed Ibrahim Dr. Lamia Hassaan

B

Academic Year: 2023-2024
Term: 1st Mid-Term (Credit)
Subject: Data Structures
Code: CS201
Specialization: 2nd Level CS
Aids: None
Total Marks: 20

Number of Questions: 2 **Model Answer** **Number of pages: 3**

Model Answer B 1/3

Answer the following questions:

First Question: **(5 Marks (1 Mark each))**

Fill in the spaces with suitable word:

a. **enqueue():** inserts an element at the **rear** of the **Queue**.
b. The disadvantage of Linked List is **Linear Access** **and** **Extra Memory**
c. Function **pop ()** in **Stack** used to **delete item**

Show which sentence is true and which is false and correct the false sentences:

d. Stack is a data structure which insertions and deletions are restricted to one end. (**True**)
e. A Stack is called a FIFO list. (**False**) **FILO**

Second Question: Write C++ code to implement the following functions:
(15 Marks (5 Marks each))

a- **Push** and **IsFull** Stack functions.

Answer:

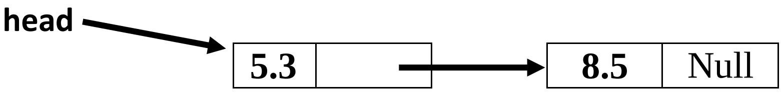
```
//check if stack is full
bool isFull(int top)
{
   if (top == MAXSIZE - 1) return true;
   else return false;
}
//Push function
void push(int item)
{
if (isFull(top))
cout<<"Stack overflow";
else
{  top = top +1;   stack_arr[top]=item; }
}
```

b- **Dequeue** and **IsEmpty** Queue functions.

<u>Answer:</u>

```
bool isEmpty(int head)
{
        if (head== -1|| head>tail)  return true;
        else return false;
}
int dequeue()
        {       int element;
                if (isEmpty(head))
                {
                 cout<<"Queue Underflow ";
                 return -1;
                }
                else {
                        element= queue[head] ;
                        head++;
                 }

                return element;
        }
```

c- Write **node class**, **linked list class**, **insertion <u>at the front</u>** function then use them to create the linked list as the following figure (**insert value 8.5 then value 5.3**).



<u>Answer:</u>

```
#include <stdafx.h>
#include <iostream>
#include <string>
using namespace std;
class Node
{
public:
        float data;
        Node *next;
        Node()
        {
                data=0;
                next=NULL;
        }
```

```
};
```

Model Answer B 3/3

```
class Linkedlist
{
public:
        Node *head;
        Linkedlist()
        {
                head=NULL;
        }

        bool isEmpty()
        {
                if (head==NULL)
                        return true;
                else
                        return false;
        }
        void insertatbeginning(int item)
        {
                Node *newnode=new Node();
                newnode->data=item;
                if(isEmpty())
                {
                        newnode->next=NULL;
                        head=newnode;
                }
                else
                {
                        newnode->next=head;
                        head=newnode;
                }
        }
};
void main()
{
        Linkedlist list;
        list.insertatend(8.5);
        list.insertatend(5.3);
}
```

With my best wishes

Ministry for High Education
Modern Academy for Computer Science and Management Technology
Department: Computer Science
Program: Computer Science
Time: 1 Hrs.
Examiners: Dr. Ahmed Ibrahim Dr. Lamia Hassaan

Academic Year: 2023-2024
Term: 1st Mid-Term (Credit)
Subject: Data Structures
Code: CS201
Specialization: 2nd Level CS
Aids: None
Total Marks: 20

A

Number of Questions: 2	Model Answer	Number of pages: 3

Answer the following questions:

First Question: **(5 Marks (1 Mark each))**

Fill in the spaces with suitable word:

a. Removing an element occurs at the **front** of the **Queue**

b. The disadvantage of **array** is **fixed size**

c. Function **push ()** in Stack used to **insert item**

Show which sentence is true and which is false and correct the false sentences:

d. Queue is a data structure which insertions and deletions are restricted to one end. (**False**) **insertions at rear and deletions at front**

e. A Stack, in other words, is called a LIFO list. (**True**)

Second Question: Write C++ code to implement the following functions:
(15 Marks (5 Marks each))

a- **POP** and **IsEmpty** Stack functions.

Answer:

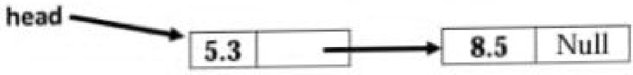
```
//check if stack is empty
bool isEmpty(int top)
{
   if (top == -1) return true;
   else return false;
}
//Pop function
int pop( )
{
if (isEmpty(top)) { cout<<"Stack underflow"; return -1;}
else
{ int item = stack_arr[top]; top = top-1; return item; }
}
```


b- **Enqueue** and **IsFull** Queue functions.

Answer:

```
bool isFull(int tail)
{
      if ( tail== maxsize-1)
            return true;
      else return false;
}
void enqueue(int item)
      {
            if(isFull(tail))
                  { cout<<"\nQueue is full"; return;}
            else
                  {
                        if ( head == -1 ) head = 0;
                        tail = tail + 1;
                        queue[tail] = item;
                  }
            }
```

c- Write **node class**, **linked list class**, **insertion** <u>**at the front**</u> function then use them to create the linked list as the following figure (**insert value 8.5 then value 5.3**).



<u>Answer:</u>

```
#include <stdafx.h>
#include <iostream>
#include <string>
using namespace std;
class Node
{
public:
        float data;
        Node *next;
        Node()
        {
                data=0;
                next=NULL;
        }
```

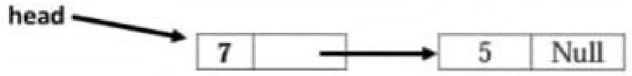
```
};
```

Model Answer B 3/3

```
class Linkedlist
{
public:
        Node *head;
        Linkedlist()
        {
                head=NULL;
        }

        bool isEmpty()
        {
                if (head==NULL)
                        return true;
                else
                        return false;
        }
        void insertatbeginning(int item)
        {
                Node *newnode=new Node();
                newnode->data=item;
                if(isEmpty())
                {
                        newnode->next=NULL;
                        head=newnode;
                }
                else
                {
                        newnode->next=head;
                        head=newnode;
                }
        }
};
void main()
{
        Linkedlist list;
        list.insertatend(8.5);
        list.insertatend(5.3);
}
```

With my best wishes

c- Write **node class**, **linked list class**, **insertion at the end** function then use them to create the linked list as the following figure (**insert value 7 then value 5**).



Answer:

```
#include <stdafx.h>
#include <iostream>
#include <string>
using namespace std;
class Node
{
public:
        int data;
        Node *next;
        Node()
        {
                data=0;
                next=NULL;
        }
};
class Linkedlist
{
```



```
public:
        Node *head;
        Linkedlist()
        {
                head=NULL;
        }

        bool isEmpty()
        {
                if (head==NULL)
                        return true;
                else
                        return false;
        }
        void insertatend(int item)
        {
                Node *newnode=new Node();
                newnode->data=item;
                newnode->next= NULL;
                Node *temp1=head;
                while (temp1->next!=NULL) temp1=temp1->next;
                temp1->next=newnode;
        }
};
void main()
{
        Linkedlist list;
        list.insertatbeginning(7);
        list.insertatbeginning(5);
}
```

With my best wishes

Ministry for High Education
Modern Academy for Computer Science and Management Technology
Department: Computer Science
Program: Computer Science
Time: 1 Hrs.
Examiners: Dr. Ahmed Ibrahim Dr. Lamia Hassaan

B

Academic Year: 2023-2024
Term: 1^{st} Mid-Term (Credit)
Subject: Data Structures
Code: CS201
Specialization: 2^{nd} Level CS
Aids: None
Total Marks: 20

Number of Questions: 2	Model Answer	Number of pages: 3

Answer the following questions:

First Question: **(5 Marks (1 Mark each))**

Fill in the spaces with suitable word:

a. **enqueue():** inserts an element at the **rear** of the **Queue**.
b. The disadvantage of Linked List is **Linear Access** and **Extra Memory**
c. Function **pop ()** in **Stack** used to **delete item**

Show which sentence is true and which is false and correct the false sentences:

d. Stack is a data structure which insertions and deletions are restricted to one end. (**True**)
e. A Stack is called a FIFO list. (**False**) **FILO**

Second Question: Write C++ code to implement the following functions:
(15 Marks (5 Marks each))

a- **Push** and **IsFull** Stack functions.

Answer:

```
//check if stack is full
bool isFull(int top)
{
   if (top == MAXSIZE - 1) return true;
   else return false;
}
//Push function
void push(int item)
{
if (isFull(top))
cout<<"Stack overflow";
else
{ top = top +1;   stack_arr[top]=item; }
}
```


b- **Dequeue** and **IsEmpty** Queue functions.

Answer:

```
bool isEmpty(int head)
{
        if (head== -1|| head>tail)  return true;
        else return false;
}
int dequeue()
        {       int element;
                if (isEmpty(head))
                {
                 cout<<"Queue Underflow ";
                 return -1;
                }
                else {
                        element= queue[head] ;
                        head++;
                 }

                return element;
        }
```